CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 10: **Tissues in Action** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Tissue between the vertebrae that acts as a cushion: 1

(a) cartilage **(b)** blood **(c)** muscle **(d)** bone

2. Bark consists of: 1

(a) xylem **(b)** cortex only **(c)** cork and phloem **(d)** pith

3. Which of these tissues has sieve plates? 1

(a) xylem **(b)** cork **(c)** phloem **(d)** collenchyma

4. The matrix in blood is: 1

(a) plasma **(b)** fat **(c)** cellulose **(d)** calcium salts

5. Fat stored below the skin gives: 1

(a) colour **(b)** insulation **(c)** speed **(d)** hardness

6. Plants cannot move but animals can. This is related to the presence of: 1

(a) xylem
(b) phloem
(c) supportive tissues in plants and muscular tissue in animals
(d) stomata

7. The dead tissue that conducts water is: 1

(a) collenchyma **(b)** sieve tubes **(c)** xylem vessels **(d)** companion cells

8. RBCs carry oxygen with the help of: 1

(a) WBC **(b)** platelets **(c)** haemoglobin **(d)** plasma

9. Ligaments are: 1

(a) elastic and connect bones **(b)** tough and connect muscles to bones
(c) nervous **(d)** fluid

10. A cut made on the stem of a plant that removes a ring of bark will prevent the movement of: 1

(a) light **(b)** gases **(c)** water upward **(d)** food downward

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is the cardiac muscle called involuntary and striated? Give one more property. 2

12. How do the smooth muscles differ from skeletal muscles in structure and function? 2

13. Why is a woody plant able to grow thicker with time? 2

14. Name the epithelium that lines the following: 2
(a) blood vessels
(b) small intestine
(c) windpipe.

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why xylem and phloem are called complex tissues and vascular tissues. 3

16. Describe the tissues in a leaf and their functions. 3

17. Explain how a neuron transmits an impulse, in terms of the direction of the impulse. 3

OR

Compare skeletal, smooth and cardiac muscles in a table of three points. 3

18. Explain why the healing of a wound is faster in some tissues than others. 3

OR

Explain how a plant gets support without bones. Name the tissues and the structures. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the types of epithelial and connective tissues and describe how their structure is related to function with five examples. 5

OR

Describe the types of joints, the tissues involved (bone, cartilage, ligament, synovial fluid) and the movement permitted by each, with a table. 5

20. Compare the working of voluntary and involuntary muscles in the body, with examples and the control by the nervous system. 5

OR

Explain the structure and function of blood: plasma, RBC, WBC and platelets, with the numbers or the key features. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND